

THE TRAINER

v0.1 Freeze and Forensic Dossier

Smoke-validated functional baseline

Validated source	90196251d59907754cc19caf76b59a737752893b
Validated image	sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89
Audit source HEAD	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Generated	2026-06-30T04:24:38Z

Evidence only. No runtime rebuild, GPU retraining, or stable tag mutation was performed.

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1. Executive summary

The Trainer v0.1 has one evidence-backed end-to-end smoke run: Prueba02. That run received a Surveyor scene package, verified its checksum, accepted the COLMAP scene, trained on CUDA for 300 steps, wrote `ckpt_299_rank0.pt`, exported standard and compressed PLY, packaged and transferred the result, and loaded the standard PLY in SuperSplat v2.27.4.

The immutable runtime identity that was published during Prueba02 is:

- source commit: 90196251d59907754cc19caf76b59a737752893b;
- source tree: 9568224839eb12bfd0bfbca4625bdea3bfe7a8bb;
- workflow: Publish Headless gsplat Dev, run 28285526569, successful;
- published development tag:
`docker.io/${DOCKERHUB_USERNAME}/cloud-workstation:headless-gsplat-v0.1-dev`;
- published manifest digest:
`sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89`.

This digest is identified by workflow chronology: it was the last successful publication before the 2026-06-29 Prueba02 validation and remained the development tag target throughout the run. The pod-side image digest was not captured in the preserved Prueba02 transcript, so the linkage is a strong forensic inference rather than a directly recorded pod observation.

The read-only source branch was later advanced to:

- source HEAD: 221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3;
- source tree: f26a1a2fddd2f0a3398b3b49b5fbb61ea24038d5;
- current publication digest:
`sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98`.

That later image adds the Trainer-side preparar -escena command and its regression suite. It was not the image used for Prueba02 and must not replace the validated digest in the stable v0.1 tag proposal.

Proposed immutable identifiers:

- Git tag `trainer-v0.1.0-smoke-validated` targeting commit
`90196251d59907754cc19caf76b59a737752893b`;
- Docker tag `headless-gsplat-v0.1.0` targeting digest
`sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89` without rebuilding.

No Git tag or Docker tag was created by this audit.

2. Evidence language and confidence

This dossier uses three evidence levels:

- **Validated**: supported by versioned evidence, a preserved workflow log, an executable local test, or an inspected artifact.
- **Reported**: preserved as an operator observation, but the original raw artifact or log is unavailable.
- **Inferred**: derived from immutable chronology or source behavior and explicitly identified as inference.

This distinction is important because the repository preserves the Prueba02 report and screenshot, but not the large scene archive, checkpoint, PLY files, full pod transcript, or pod-side image digest.

3. Git identity and chain of custody

Item	Value
Remote fetch/push	https://github.com/Marquezmemo/cloud-workstation.git
Source branch	headless-gsplat-v0.1-dev
Source HEAD recorded before audit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Source tree	f26a1a2fddd2f0a3398b3b49b5fbb61ea24038d5
Audit branch	audit/trainer-v0.1-freeze
Audit branch base	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Validated image source commit	90196251d59907754cc19caf76b59a737752893b
Validated image source tree	9568224839eb12bfd0bfbc4625bdea3bfe7a8bb

The tracked source tree had no diff at audit start. The original shared worktree contained an unrelated untracked output/ directory with Surveyor freeze artifacts. It was not modified, moved, staged, or copied. The audit was created in an isolated worktree based on the exact source HEAD.

The audit changes only this dossier and its generated artifacts. Functional paths such as Dockerfile, workflows, requirements, scripts, tests, trainer, patch, exporter, and packaging remain byte-for-byte equal to source HEAD 221c4ef.

4. Docker identity

4.1 Validated Prueba02 image

Item	Value
Workflow	Publish Headless gsplat Dev
Workflow run	https://github.com/Marquezmemo/cloud-workstation/actions/runs/28285526569
Result	success
Source commit	90196251d59907754cc19caf76b59a737752893b
Tag configured by workflow	docker.io/\${DOCKERHUB_USERNAME}/cloud-workstation:headless-gsplat-v0.1-dev
Manifest digest	sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89
Platform	linux/amd64

4.2 Current development publication

Item	Value
Workflow run	https://github.com/Marquezmemo/cloud-workstation/actions/runs/28413758338
Source commit	221c4ef76a1ff5bfa8a9f45e5e9a084e8fbedfe3
Manifest digest	sha256:726ac98c9678431fc34fbcc1de0bf07b71adbff215d672f04740a95999a6bf98
Relationship to Prueba02	post-validation image; not used by Prueba02

The failed documentation-triggered workflow run 28411332932 did not replace the development tag. The next successful run was the post-validation 221c4ef build.

4.3 Base image

```
runpod/pytorch:2.4.0-py3.11-cuda12.4.1-devel-ubuntu22.04
@sha256:61a4aafb0094cd773f11eefa378929d5a687bd775febeb78eac62fc824141fb5
```

The base image is pinned by manifest digest. The Trainer does not use latest.

5. Runtime and fixed dependencies

5.1 Runtime identity

Component	Identity	Evidence note
Ubuntu	22.04 base	Dockerfile/base identity
Python	3.11 line; 3.11.10 previously observed	Earlier RunPod runtime evidence; not separately recaptured in the Prueba02 record
PyTorch	base 2.4.0 line; 2.4.1+cu124 previously observed	Earlier RunPod runtime evidence
CUDA toolkit/runtime in Trainer	12.4 line	Base tag and earlier runtime evidence
GPU used by Prueba02	NVIDIA GeForce RTX 4090	Operator and validation record
gsplat	1.5.3	Exact requirement and build label
OpenCV	opencv-python-headless==4.10.0.84	Exact requirement
runpodctl	2.5.0	Docker build ARG and verified binary checksum

The Surveyor runtime had a different CUDA stack. Its reported CUDA versions do not redefine the Trainer image identity.

5.2 Core Python requirements

```
gsplat==1.5.3
jaxtyping==0.3.11
ninja==1.13.0
numpy==1.26.3
rich==14.3.4
typing_extensions==4.15.0
wadler-lindig==0.1.7
```

5.3 Official trainer requirements

```
fused-ssim @
git+https://github.com/rahul-goel/fused-ssim@328dc9836f513d00c4b5bc38fe30478b4435cbb5
imageio[ffmpeg]==2.37.3
matplotlib==3.11.0
numpy==1.26.3
opencv-python-headless==4.10.0.84
piexif==1.1.3
pycolmap @ git+https://github.com/rmbruaalla/pycolmap@cc7ea4b7301720ac29287dbe450952511b32125e
pyyaml==6.0.2
scikit-learn==1.4.2
scipy==1.12.0
tensorboard==2.20.0
torchmetrics[image]==1.9.0
tqdm==4.68.2
tyro==1.0.13
```

Direct dependencies are pinned, including both Git dependencies by commit. Transitive dependencies are not fully locked with hashes. Historical pip-freeze.txt and dpkg-freeze.txt belong to the earlier workstation baseline and are not treated as a complete freeze of the Prueba02 Trainer image.

5.4 System packages added by Trainer

```
git cmake ninja-build build-essential ffmpeg wget curl unzip nano htop tmux ca-certificates
```

6. Official source and exact headless patch

The training backend is the official gsplat example:

```
repository: https://github.com/nerfstudio-project/gsplat
release line: v1.5.3
commit: 937e29912570c372bed6747a5c9bf85fed877bae
path: examples/simple_trainer.py
installed path: /opt/gsplat/examples/simple_trainer.py
```

The build removes the cloned .git directory after checking out the fixed commit.

patch-gsplat-simple-trainer-headless.sh performs only these source substitutions:

1. Replaces `import viser` with `viser = None` and a headless comment.
2. Replaces `GsplatViewer` and `GsplatRenderTabState` imports with local stubs. Constructing the viewer stub raises an explicit headless error.
3. Replaces `CameraState`, `RenderTabState`, and `apply_float_colormap` imports with object/function stubs. Calling the colormap stub raises an explicit headless error.
4. Verifies that the three top-level viewer import forms no longer remain.

The patch does not alter training math, camera parsing, optimizer behavior, checkpoint serialization, PLY export, or dataset loading. The wrapper also passes `--disable_viewer` and rejects `DISABLE_VIEWER=false`.

7. Training architecture

```
Surveyor package
-> /workspace/scenes/<scene>
-> train-scene.sh --check <scene>
-> official simple_trainer.py default --disable_viewer
-> CUDA training
-> /workspace/outputs/<scene>/ckpts/ckpt_<step>_rank<rank>.pt
-> generar / generate-ply.py
-> standard and compressed PLY
-> empaquetar
-> runpodctl transfer
-> independent inspection / SuperSplat
```

The official trainer initializes CUDA device `cuda:<local_rank>`, parses COLMAP data, creates splats and optimizers, trains for `max_steps`, and writes the final checkpoint at `max_steps - 1`. This explains `MAX_STEPS=300` producing `ckpt_299_rank0.pt`.

The checkpoint contains at least:

```
step
splats: means, scales, quats, opacities, sh0, shN
```

Optional pose or appearance state is added only when the corresponding features are enabled.

8. Workspace, command surface, and persistence

8.1 Validated image layout at commit 9019625

```
/workspace/datasets
/workspace/scenes
/workspace/outputs
/workspace/logs
/workspace/checkpoints
```

8.2 Current source HEAD addition

Commit 221c4ef additionally creates:

```
/workspace/temp
```

Persistence is not provided by directory creation alone. Datasets, scenes, outputs, logs, checkpoints, and transfer artifacts are persistent only when `/workspace` is backed by RunPod persistent storage or a mounted volume.

8.3 CMD

```
CMD ["runpod-keepalive.sh"]
```

The `keepalive` command creates `/workspace/logs`, prints validation hints, and runs `sleep infinity` so a non-interactive pod remains available.

8.4 Commands

Command	Responsibility	Prueba02 image
<code>prepare-dataset.sh</code>	Creates an empty COLMAP scene skeleton	present
<code>train-scene.sh --check</code>	Structural dataset gate and run metadata	present and exercised
<code>train-scene.sh</code>	Wraps official trainer, logs and checkpoint alias	present and exercised
<code>generar / Generar</code>	Calls <code>generate-ply.py</code>	present and exercised
<code>empaquetar</code>	Archives PLY exports and writes checksum	present and exercised
<code>comprimir</code>	Alias for <code>empaquetar</code>	present
<code>empaquetar -logs</code>	Packages scene-scoped existing evidence with portable checksum	present; not part of preserved Prueba02 sequence
<code>collect-training-diagnostics.sh</code>	Captures runtime, GPU, environment, trees and recent logs	present
<code>validate-gpu.sh</code>	Verifies NVIDIA, torch, CUDA operation and <code>gsplat</code> import	present
<code>preparar-escena</code>	Securely verifies and imports Surveyor archive	added at 221c4ef after Prueba02

9. Surveyor-to-Trainer scene contract

Prueba02 arrived as `Prueba02-surveyor-scene.tar.gz` with an adjacent portable checksum record. The checksum was verified in Surveyor and after receipt by Trainer. The exact scene archive SHA-256 was not preserved in the attachments and is not reconstructed here.

Minimum Trainer scene structure:

```
/workspace/scenes/Prueba02/
  images/
  sparse/0/cameras.bin
  sparse/0/images.bin
  sparse/0/points3D.bin
  surveyor-manifest.json
```

Surveyor evidence records 30 `input/database/registered` images, one camera, `camera_model=OPENCV`, two sparse candidates, and selection of the 30-image, 4,555-point model normalized to `sparse/0`.

During Prueba02 the archive was manually extracted before `train-scene.sh --check` passed.

`preparar-escena` automates that segment in current source HEAD, including checksum verification, safe tar inspection, manifest consistency, transactional overwrite and rollback. It is regression-tested but was not available in the validated Prueba02 image.

10. Camera and real OPENCV undistortion behavior

Prueba02 used one COLMAP OPENCV camera. The fixed official parser does not ignore distortion.

For every camera it reads:

```
fx, fy, cx, cy
```

For OPENCV it also reads:

```
k1, k2, p1, p2
```

It constructs the initial intrinsic matrix K , then for perspective distortion:

1. Calls `cv2.getOptimalNewCameraMatrix(K, params, size, 0)`.
2. Calls `cv2.initUndistortRectifyMap(...)` with the corrected matrix.
3. Stores the corrected intrinsic matrix and valid ROI.
4. Updates the effective image size to the ROI dimensions.
5. In dataset loading, calls `cv2.remap(...)`.
6. Crops the remapped image to the valid ROI.
7. Supplies the corrected image and corrected K to training.

The warning `COLMAP Camera is not PINHOLE. Images have distortion.` reports the input model. It is not evidence that distortion is ignored.

Surveyor copies source images into the scene and runs feature extraction, matching, and mapping. It does not invoke `COLMAP image_undistorter`. Therefore v0.1 performs one undistortion in the fixed `gsplat` parser. It does not change the Surveyor camera to `PINHOLE`, and it does not perform a second undistortion.

Double-undistortion risk: a future implementation must change both halves of the contract together. If Surveyor starts delivering already-undistorted images but leaves OPENCV distortion parameters active, the Trainer parser will remap them again. If Surveyor changes the camera metadata without matching image geometry and corrected intrinsics, training will receive an inconsistent camera model.

Primary source:

```
https://github.com/nerfstudio-project/gsplat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/datasets/colmap.py
```

11. Checkpoints, outputs, and exports

`train-scene.sh` writes:

```
/workspace/logs/<scene>/run.env
/workspace/logs/<scene>/run.summary
/workspace/logs/<scene>/train.log
/workspace/logs/<scene>/error.tail      # failure only
/workspace/outputs/<scene>/...
/workspace/checkpoints/<scene>        # symlink to outputs/<scene>/ckpts
```

The official trainer creates:

```
ckpts/
stats/
renders/
ply/
tb/
```

`generate-ply.py` discovers or accepts an explicit checkpoint, loads tensors, normalizes required fields, uses `gsplat.export_splats`, and writes:

```
/workspace/outputs/<scene>/exports/<scene>.ply
/workspace/outputs/<scene>/exports/<scene>.compressed.ply
/workspace/logs/<scene>/generate-ply.log
```


The standard PLY gate requires binary little endian encoding and a vertex count equal to the checkpoint splat count. The compressed gate rejects a count larger than the input count.

empaquetar includes existing .ply files in <scene>-ply-exports.tar.gz. Its checksum record contains the absolute pod path in the validated v0.1 behavior. It can be checked in place; after transfer the digest must be compared manually. empaquetar - logs uses a relative, portable checksum but does not generate telemetry.

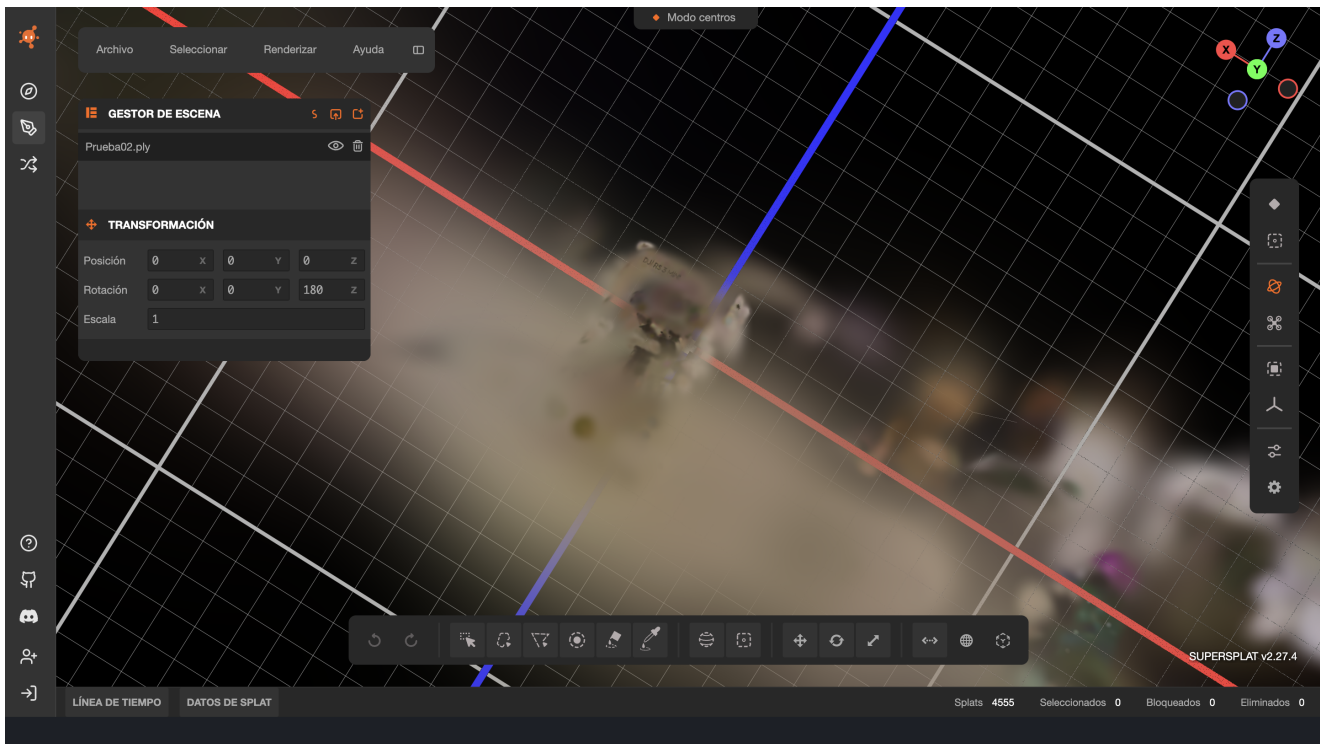
12. Prueba02 evidence

Observation	Result	Evidence level
Scene archive	Prueba02-surveyor-scene.tar.gz; checksum passed	Validated; exact archive digest not preserved
Scene archive size	741.1 MB; transfer UI showed 777 MB	Reported and versioned in Surveyor record
Images	30	Validated
Camera	1 x OPENCV	Validated
Training depth	MAX_STEPS=300	Validated
Duration	2:20	Validated record
Throughput	2.13 it/s	Validated record
Final reported loss	0.196	Reproducibility observation only; not a quality score
Initial Gaussians	4,555	Validated record
Final Gaussians	4,555	Validated record
Checkpoint	ckpt_299_rank0.pt	Validated record
Reported GPU memory	3.2356 GB	Validated record
Standard PLY	1,076,455 bytes	Validated record
Compressed PLY	280,894 bytes	Validated record
Standard vertices	4,555	Independently inspected
Compressed vertices	4,555	Export log/operator record
Standard non-finite values	0 NaN, 0 infinities	Independently inspected
Compressed non-finite values	0 NaN, 0 infinities	Operator assertion in freeze mission; raw compressed artifact/inspection output not versioned
SuperSplat	Standard PLY loaded and rendered as 4,555 splats in v2.27.4	Validated screenshot

The independently inspected standard PLY had SHA-256:

```
f7acbb332667a028756f7f2426dd07bfde577f66f30ef0dfa27b02449eea5c71
```

It was binary little endian, contained 59 float properties, had an exact payload for its header and vertex count, and contained no NaN or infinite values.



Prueba02 loaded in SuperSplat

The unchanged Gaussian count means the run did not validate densification. The screenshot proves compatibility and rendering, not professional geometry, commercial quality, capture quality, or production readiness.

13. Tests executed for this freeze

No GPU training, model download, image build, or image publication was performed.

Non-destructive checks executed from the audit workflow:

```
bash -n for all existing Trainer shell entrypoints
bash tests/test-preparar-escena-trainer.sh
git diff --exit-code <source-head> -- Dockerfile workflows requirements scripts tests
```

Result:

```
All 15 preparar-escena tests passed.
Functional diff against source HEAD: zero.
```

The suite covers help/arguments, success through the productive `train-scene.sh --check`, missing files, checksum mismatch, path traversal, absolute paths, multiple roots, filename/root mismatch, invalid manifest, manifest consistency, COLMAP structure, existing destination, safe overwrite, rollback, capitalization, and source archive preservation.

These tests validate current source HEAD 221c4ef. The validated Prueba02 image at commit 9019625 predates `preparar-escena` and its suite.

14. Limitations

- Densification was not validated; the Gaussian count stayed at 4,555.
- Professional quality was not validated.
- Commercial quality was not validated.
- The loss value 0.196 is not a quality score.
- PSNR, SSIM, LPIPS, comparable renders, and a quality threshold were not preserved.
- The full scene archive, checkpoint, PLY files, compressed PLY inspector output, and full pod logs are outside Git.

- The exact scene archive digest was not preserved in the local attachments.
- The pod-side image digest was not captured. The validated digest association is based on immutable workflow chronology.
- `empaquetar` writes an absolute-path checksum in the validated image.
- GPU telemetry was not continuously packaged by Trainer.
- Files above 1 GB, interrupted transfers, resume behavior, and measured transfer throughput remain unvalidated.
- The first run downloaded a 233 MB AlexNet model and spent 61.61 seconds setting up a CUDA extension; these are operational observations, not frozen performance guarantees.

15. Risks

1. **Moving development tag.** `headless-gsplat-v0.1-dev` is mutable and has already moved beyond the `Prueba02` image.
2. **Source/image mismatch.** Tagging current HEAD as if it produced the validated digest would be incorrect.
3. **Unpinned transitive dependencies.** Rebuilding the same Dockerfile can produce a different manifest digest.
4. **Double undistortion.** Changing only Surveyor image geometry or only camera metadata can cause a second remap or inconsistent intrinsics.
5. **Evidence gaps.** Missing large artifacts prevent independent replay of every numerical claim.
6. **Short smoke depth.** A 300-step run proves pipeline function, not convergence or final quality.
7. **Diagnostics sensitivity.** Global diagnostics can contain environment data and must be handled as potentially sensitive evidence.

16. Recovery procedure

To recover the exact validated runtime without rebuilding:

1. Resolve the proposed Git tag target to commit 90196251d59907754cc19caf76b59a737752893b.
2. Pull the image by digest, not by mutable development tag:
`docker.io/${DOCKERHUB_USERNAME}/cloud-workstation@sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89`
3. Mount persistent storage at /workspace.
4. Restore Prueba02-surveyor-scene.tar.gz and its adjacent checksum.
5. Verify the checksum from the archive directory.
6. Because the validated image predates preparar-escena, extract manually into /workspace/scenes.
7. Run `train-scene.sh --check Prueba02`.
8. For a pipeline smoke replay, use the preserved command `depth MAX_STEPS=300`. This is an optional replay procedure, not an action performed by this audit.
9. Export with `generar --scene Prueba02`, package with `empaquetar Prueba02`, and compare artifacts against preserved sizes/counts where available.

To create the proposed stable Docker tag without rebuilding, an authorized registry operation should copy the existing digest to `headless-gsp-lat-v0.1.0`. The operation must verify the resulting tag resolves to the same manifest digest. This audit does not execute that mutation.

17. Handoff to The Eye

The Eye should integrate the following facts into official documentation without altering runtime behavior:

- distinguish source HEAD 221c4ef from validated runtime source 9019625;
- record validated digest `sha256:535822...b63a89` and current post-validation digest `sha256:726ac98...a6bf98` separately;
- record the OPENCV remap/crop behavior and remove any implication that the warning means distortion is ignored;
- preserve the double-undistortion warning;
- state that `preparar-escena` is post-Prueba02 functionality validated by regression/CI, not by the Prueba02 pod run;
- preserve Prueba02 numbers with evidence grades and limitations;
- do not claim densification, professional quality, or commercial quality;
- do not claim the compressed PLY non-finite scan is independently reproducible unless its raw inspection evidence is preserved;
- adopt the proposed Git and Docker tags only after verifying their exact targets;
- retain this report, JSON manifest, PDF, and SHA256SUMS as one evidence set.

18. Primary references

- Repository: <https://github.com/Marquezmemo/cloud-workstation>
- Validated image workflow:
<https://github.com/Marquezmemo/cloud-workstation/actions/runs/28285526569>
- Current source workflow:
<https://github.com/Marquezmemo/cloud-workstation/actions/runs/28413758338>

- Official trainer: https://github.com/nerfstudio-project/gspat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/simple_trainer.py
- Official COLMAP parser: <https://github.com/nerfstudio-project/gspat/blob/937e29912570c372bed6747a5c9bf85fed877bae/examples/datasets/colmap.py>
- Versioned Prueba02 record: docs/validation-runs/2026-06-29-prueba02-trainer-e2e.md
- Versioned SuperSplat evidence: docs/evidence/prueba02-supersplat.png

19. Freeze declaration

The Trainer v0.1 smoke baseline is functionally demonstrated by Prueba02 under the immutable image digest sha256:535822530f4b23fca3eee3970b147e3e979162a32c2e9de9f76a829836b63a89. The audit branch adds evidence only. Functional diff against source HEAD is zero.

This is a smoke-validated baseline, not a quality-certified release.